Adapting RadarHD to SLAM-RF: Deep Learning for Radar-Based Spatial Reconstruction

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Abstract

Millimeter-wave radar is a robust sensing modality for robotic perception under adverse environmental conditions, but its limited angular resolution and characteristic artifacts hinder its direct use for high-fidelity mapping. RadarHD has shown that deep learning can bridge this gap by reconstructing LiDAR-like point clouds from low-resolution radar data. In this work, we adapt the RadarHD pipeline to SLAM-RF, a proprietary EPFL dataset acquired with a radar providing significantly higher azimuthal resolution than the original setting. This change introduces both architectural and computational challenges, including "non power of two" input dimensions and a substantial increase in data density.

We present the necessary modifications to the preprocessing pipeline and U-Net–based architecture to accommodate the new resolution, as well as a compute-aware experimental protocol executed on the EPFL RCP cluster. Model selection is performed through phased baseline testing and cross-validation under strict computational constraints. Quantitative evaluation using Chamfer and Modified Hausdorff distances shows that the adapted model consistently outperforms classical CFAR-based baselines and achieves performance comparable to RadarHD on the original dataset. While the higher azimuthal resolution leads to a slight improvement in median Chamfer distance, it does not yield a consistent gain in Hausdorff distance, highlighting the non-trivial interplay between input resolution, noise, and generalization. These findings suggest promising directions for future work, particularly in preprocessing optimization and architectural refinement.

1 Introduction

LiDAR is often treated as a reference sensor for mapping, localization, and collision avoidance in robotics because it can produce dense, low-noise point clouds that are straightforward to associate across successive viewpoints. Despite this widespread use, LiDAR can degrade severely in the presence of occlusions and strong scattering, for instance when robots operate in heavy fog, smoke, or dust, as in search-and-rescue, disaster response, and firefighting. These scenarios call for sensing modalities that can penetrate such obscurants while still providing reliable scene structure. Radar, the radio-frequency counterpart of LiDAR, is attractive in this respect since RF (which stands for Radio Frequency) propagation is more resilient to these conditions. The trade-off is resolution: due to longer RF wavelengths, single-chip radars typically achieve angular resolution that is on the order of two magnitudes lower than LiDAR, yielding much coarser point clouds and restricting them to lower-fidelity tasks such as basic collision avoidance. Higher-resolution use cases often rely on large mechanically scanned radar systems, which increases both size and cost.

In [2], the authors propose RadarHD, a tailored end-to-end neural network that reconstructs LiDAR-like point clouds from low-resolution radar point clouds. This model shows strong performance on scene reconstruction, odometry, and mapping, and remains effective even in smoke-filled environments. RadarHD addresses a broader and more general task: generating high-resolution point clouds directly from radar heatmaps, with quality comparable to LiDAR outputs.

Our goal in this project is to adapt the codebase from [2] to operate on a higher-resolution dataset, specifically SLAM-RF. The SLAM-RF dataset consists of around 10k pairs of radar, lidar measurements taken both indoors and outdoors around the EPFL campus in unoccluded environments. All of this is described in the project specification [1]. Crucially, this dataset provides us with higher resolution measurements than the ones used by original researchers. Our interest lies in evaluating the performance of a U-Net type architecture on better data.

Our contribution

Our main objective is to adapt the RadarHD codebase to the SLAM dataset, which presents several challenges. Radar–LiDAR synchronization is non-trivial due to the radar acquisition pipeline and frame structure, and converting raw I/Q data into polar range–azimuth images involves multiple design choices. A key difference from the original RadarHD pipeline lies in azimuth processing: while RadarHD uses a spatial Fourier transform to produce 64 angular bins, our SLAM dataset provides 180 beamformed angles. This mismatch in azimuth resolution motivates our study and requires both data preprocessing and architectural adaptations. In particular, we must downscale the input along a single axis and handle a non–power-of-two number of azimuth bins while preserving the U-Net structure. Finally, we postprocess the outputs to ensure fair comparison with prior work, evaluating whether higher azimuth resolution improves performance and reduces reflection-induced artifacts observed in the original RadarHD setting.

2 Methods

2.1 Dataset preparation

Radar measurements are well known to be affected by spurious artifacts, such as sidelobes generated by strong reflectors, which produce sinc-like patterns along the azimuth dimension. These effects persist even in higher-resolution radar systems. A key design choice was therefore whether to feed the model with completely raw measurements or with data that had undergone limited processing to acquire a spatial interpretation. While, in principle, a deep neural network could learn operations such as the FFT directly from raw inputs, we followed the approach of the original authors and opted for beamformed data. This choice provides inputs with a clear physical meaning, facilitates interpretability, and enables image enhancement while preserving the underlying radar physics.

2.2 Preprocessing

Both the SLAM-RF and RadarHD datasets provide paired radar LiDAR measurements acquired in comparable indoor environments. RadarHD constructs range–azimuth heatmaps by applying a 2D FFT to raw complex I/Q samples. In our setting, the higher angular resolution of the SLAM-RF radar enables an equivalent range–azimuth representation obtained via beamforming, producing polar heatmaps of size 512×180 (**range** $\times$ **azimuth**), as opposed to the 512×64 Fourier-domain heatmaps used in RadarHD. Unless stated otherwise, we follow the remaining RadarHD preprocessing steps unchanged.

For radar, we apply magnitude thresholding at 0.05 and clip measurements to a maximum range of 10 m. LiDAR point clouds are cropped to a forward-facing region spanning 10 m on either side laterally, 10 m in front, and 2×0.3 m in elevation. To obtain a 2D supervisory signal consistent with the planar range–azimuth raster, we collapse the LiDAR returns across elevation by retaining, for each (r, θ) bin, the maximum intensity over all elevation angles. This aggregation is preferable to collapsing purely along z: spurious radar detections at fixed (r, θ) frequently originate from structures at different elevations but identical polar coordinates, whereas a z-centric aggregation would map such returns to different ranges and therefore entangle them with unrelated geometry.

After synchronization and labeling, the resulting dataset comprises multiple days and experiments; however, samples must still be reorganized into training examples that are consistent with the model's input/output pairing. We therefore construct train/test splits at the *experiment* level (rather than by day) and aggregate samples by sensor stream (radar or LiDAR) instead of by raw index. This choice is motivated by the fact that multiple experiments conducted on the same day often share highly similar layouts; using experiments as the smallest split unit reduces environment leakage and yields a more uniform distribution of scene configurations across training and testing. Finally, when enhanced variants of the same underlying measurement are present, we keep all associated variants grouped within the same split to prevent near-duplicate leakage and to ensure that augmentation consistency is preserved during training.

2.3 Neural network

As previously mentioned, we take as our starting point the network introduced in [2], which relies on a U-Net–based encoder–decoder architecture. This design enables multi-scale feature extraction, combining global scene understanding with precise spatial upsampling. To adjust it to the SLAM-RF dataset, we had to find a trade-off between preserving as much of the existing neural network as possible, so as not to alter the underlying process, and exploiting the higher resolution of the new dataset. Our final choice is to add a single layer at the beginning. This layer, like the subsequent ones, consists of a double convolution followed by a max-pooling operation. The former is a channel-preserving convolution, since we used the same number of channels as the original version. The latter performs the crucial reduction of the azimuth dimension from 180 to 64 bins. Since 180 is not a power of two, we adopt an adaptive pooling operation to achieve this target resolution. Several alternative extensions of the RadarHD architecture were also evaluated, including variants with only the additional double-convolution block or only the max-pooling stage. However, none of these configurations outperformed the proposed modification on the test set.

2.4 Training losses and optimizer

In the training phase, a linear combination of two different losses is minimized.

- **Pixel-wise loss (BCE).** We treat the ground-truth LiDAR map as a *binary* image and compare it to the network output after the final *sigmoid*. We use the *mean Binary Cross-Entropy* over all pixels. This loss acts *locally* (pixel-by-pixel), encouraging each pixel to match the expected label.
- **Dice loss (region-level.)** To obtain crisper and more continuous structures, we additionally use Dice loss, which measures *overlap* between prediction and ground truth and is therefore *not purely pixel-wise.* It couples pixels through a global (set/region) similarity term, improving sensitivity to thin structures and boundaries. For ground-truth g_i and prediction o_i over N pixels:

$$D = 1 - \frac{2 \sum_{i=1}^{N} o_i g_i}{\sum_{i=1}^{N} o_i^2 + \sum_{i=1}^{N} g_i^2}.$$

Training is performed by minimizing the combined loss using the Adam optimizer [3] The best ratio between the two is selected via cross-validation.

2.5 Validation and testing metrics

In order to evaluate our models, we consider two widely used distance-based metrics for point-set comparison. Both metrics quantify the discrepancy between the network output and the target (i.e., the LiDAR image) by measuring distances between the corresponding cartesian point clouds. As a result of this, both predictions and ground-truth targets must undergo a post-processing step to extract point sets prior to evaluation.

- **Chamfer distance**: for each point in one point set, it computes the distance to its nearest neighbor in the other set, and averages these distances (symmetrically) to obtain a global dissimilarity score for the pair.
- **Modified Hausdorff distance**: similarly relies on nearest-neighbor distances, but aggregates them using the median, yielding a measure that is less sensitive to extreme mismatches.

To reduce the influence of outliers at validation and testing level, we report the median metric value across the full test set (rather than the mean). Model selection is then performed by minimizing the sum of the median Chamfer distance and the median Modified Hausdorff distance.

3 Experimental Design and Implementation Details:

3.1 The Computational Bottleneck and Resource Management

Increasing the resolution from 64×256 to 180×256 substantially increased the amount of information in each frame, making the whole pipeline much slower to run. On the EPFL RCP cluster, preprocessing, training, and inference became time-consuming steps, and completing a full end-to-end cycle often required several days. This was more than a practical inconvenience: it directly influenced how we designed the project. We adopted a structured, phased workflow aimed at learning as much as possible from early experiments, limiting redundant trials, and keeping the most computationally expensive runs for the architectures that looked most promising.

Phase I: Architectural Downselection and Baseline Testing

To establish a robust performance baseline, we initiated an ablation-style screening process. The primary objective was to decouple intrinsic architectural performance from hyperparameter noise.

- **Baseline Configuration:** We initially evaluated five candidate architectures. To ensure comparability with the referenced work, we maintained a magnitude threshold of 0.05 and utilized the original researchers' default parameters.
- **Architectural Selection:** From this initial screening, two architectures emerged as the most competitive: the model described in Section 2.3 and a variant that removes the max-pooling operator. Our decision to proceed with these two was deliberate and strategically motivated by their divergent architectural philosophies. This duality allowed us to investigate how different feature extraction methods in the first architectural layer adapt to the increased azimuthal resolution of the proprietary EPFL SLAM dataset.

Phase II: Mitigating Bias via Parallel Cross-Validation

To reduce sensitivity to optimizer stochasticity and environment-specific bias, we performed 5-fold cross-validation for both selected architectures and a targeted hyperparameter sweep over a small, compute-feasible grid: We performed cross-validation over Adam hyperparameters $\texttt{lr} \in \{10^{-3}, 5 \cdot 10^{-4}, 10^{-4}\}$, $\beta_1 \in \{0.85, 0.90, 0.95\}$, $\beta_2 = 0.999$, and over the loss weighting between BCE and Dice, considering the ratios (0.9 BCE, 0.1 Dice) and (0.7 BCE, 0.3 Dice).

To maintain the integrity of the comparison, all auxiliary variables—including preprocessing enhancements and thresholding values—were kept static. This *ceteris paribus* approach ensured that any observed variance in performance could be strictly and scientifically attributed to the architectural response to high-resolution polar inputs, rather than fluctuating external parameters.

Phase III: Data Augmentation

After selecting the most promising architecture and optimizer configuration, we began to study the effect of physically motivated data enhancement on generalization. Specifically, we trained the selected model(s) on datasets generated with different enhancement levels (described in 1) while keeping the training budget and evaluation metrics fixed, in order to attribute any performance changes to augmentation-induced data diversity rather than to confounding factors.

Enhancement	**Multiplier**	**Operations**
`none`	×1	None
`low`	×4	LR, TB, LR+TB flips
`high`	×5	Range shifts + azimuth shifts
`vhigh`	×20	`low` + `high`
`vvhigh`	×28	Extended spatial shifts
`vvvhigh`	×36	Increased shift density

Table 1: *Overview of data enhancement strategies applied to beamformed radar heatmaps. All transformations preserve the physical interpretation of the polar representation.*

Due to the increased computational cost, it was not possible to conduct sufficiently thorough studies on this aspect; therefore, it is not included in the results

4 Results

After completing the baseline evaluation and cross-validation, the best-performing architecture was the one described in Section 2.3. Cross-validation selected the following hyperparameters: $lr = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a loss weighting of (0.9 BCE, 0.1 Dice). Table 2 summarizes the resulting performance and compares it with the results reported in [2]. Below, we present the cumulative distribution functions (CDFs) of the Chamfer and modified Hausdorff distances, together with a representative example of a prediction–target pair. All results are obtained using the best-performing model described above.

Metric	Original Dataset	SLAM Dataset
Median Chamfer	0.36	0.35
Median Hausdorff	0.24	0.34

Table 2: *Performance comparison between the original RadarHD and our code for SLAM dataset*

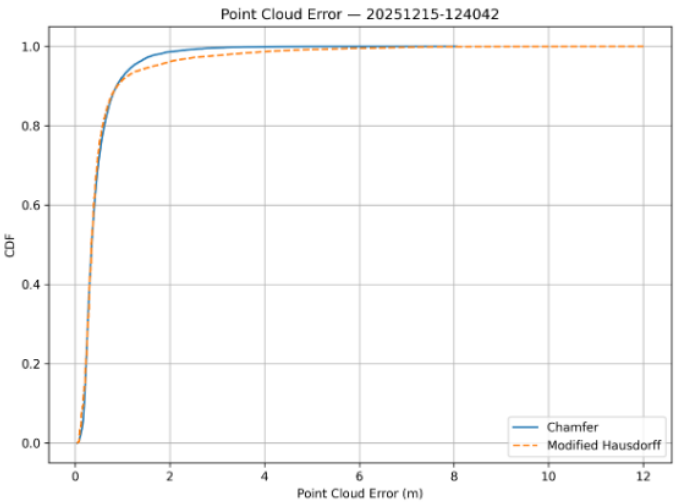

Figure 1: *Cumulative Distribution Functions of Chamfer distance and modified Hausdorff distance*

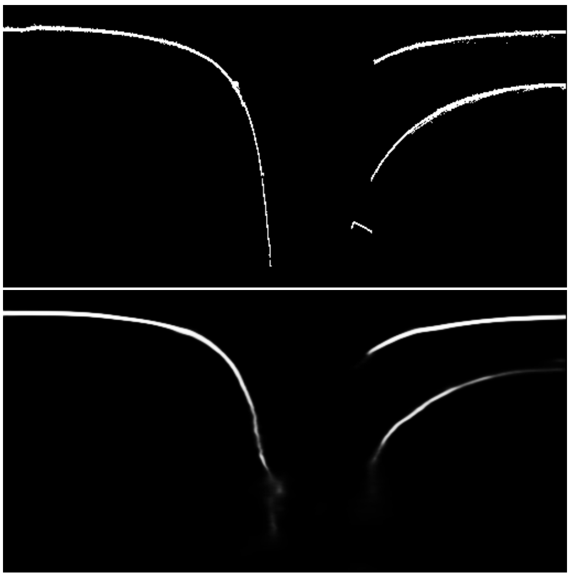

Figure 2: *Example of a prediction obtained (below) and its corresponding ground truth (above). Both images are in polar coordinates*

4.1 Range-dependent loss weighting

From qualitative inspections of outputs such as Figure 2, we observe that prediction quality degrades with increasing distance from the radar origin, i.e., at larger ranges. To mitigate this effect, we introduced range-dependent weights in the BCE term during training, defined as follows:

$$w(r) = \left(\frac{r + \varepsilon}{r_{\max}}\right)^p$$

We evaluated several values of $p \in (1, 4)$, but these trials were unsuccessful, as both distance metrics increased. For example, setting $p = 1.5$ while keeping the rest of the configuration identical to our best-performing experiment yielded a median Chamfer distance of 0.69 and a median (modified) Hausdorff distance of 0.43.

5 Discussion and Conclusions

From Table 2 and Figure 1, we observe that our model achieves a strong level of accuracy and substantially outperforms the CFAR baselines reported in [2]. When comparing our results to RadarHD on the original dataset, the median Chamfer distance improves marginally, while the median Modified Hausdorff distance shows a small degradation. These slightly inferior results were unexpected, as we initially anticipated that the increased azimuthal resolution would lead to a noticeable performance improvement. The reasons behind these results may be manifold. First, the trajectories in the new dataset differ from those in the original dataset and may exhibit lower variability, which can hinder generalization. This hypothesis is supported by an inspection of test–ground-truth image pairs, which revealed that certain trajectory patterns that are underrepresented in the dataset or present only in the testing split yield less accurate predictions. Moreover, we did not observe clear signs of overfitting, as the evaluation metrics computed on the training set are very similar to those obtained on the test set. Another possible explanation is that the preprocessing was not performed using optimal parameters (e.g., thresholding). In fact, due to the high computational cost of this procedure, we retained the parameters optimized for the original dataset, without the possibility of testing small variations that could have yielded improvements on the current dataset. Moreover, due to the high computational cost, both the baseline evaluation and the cross-validation were conducted over a limited set of architectures and hyperparameter combinations; therefore, we cannot guarantee that the selected model is globally optimal. However, we do not believe this to be the primary cause, as the different architectures did not lead to substantial performance variations and the hyperparameters selected through cross-validation coincide with those used in RadarHD. Finally, the higher azimuthal resolution, while providing richer information, may also introduce increased noise, which could in turn lead to a degradation in performance. All the reasons discussed so far suggest that our project lends itself to several directions for future exploration, including modifying the preprocessing pipeline and testing different architectures and hyperparameter settings. In addition, we implemented a data augmentation procedure within the codebase. However, experiments on the augmented dataset were limited due to the associated computational cost. Nevertheless, we believe that a thorough evaluation of data augmentation could lead to significant performance improvements.

References

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